



Time = 20 Minutes

Quiz No 7

[20 points]

Note - Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

Question 1 [10 points] Independence & Conditional Independence

The joint distribution for the two DRV's, X & Y, is given below.

Y = Second roll	4	1/20	2/20	2/20	
	3	2/20	4/20	1/20	2/20
	2		1/20	3/20	1/20
	1		1/20		
		1	2	3	4
		X = First roll			

- (a) [3 points] Are the two R.Vs X and Y independent?

Note - You need to clearly highlight the conditions for independence and justify your answer mathematically. **The points are not for the right answer but for the working around the justification of the right answer.**

- (b) [7 points] Are the two R.Vs X and Y independent, conditioned on event A, where event $A = X \leq 2$ and $Y \geq 3$?

Note - You need to clearly highlight the conditions for the independence and justify your answer mathematically. **The points are not for the right answer but for the working around the justification of the right answer.**

Question 2 [10 points] Calculating Probability for Discrete Random Variables

A Random Variable Z_n , is defined as $Z_n = \min(X_1, X_2, \dots, X_n)$, where X_i 's are outcomes of identical and independently distributed fair 6-sided die. As an example, $Z_2 = \min(X_1, X_2)$ – meaning Z_2 is the minimum of the outcome of the two independent throws of the fair 6-sided die X_1, X_2 .

- (a) [5 points] Calculate the probability that $P(1 < Z_{12} < 6)$.

- (b) [5 points] Find the minimum value of n such that $P(Z_n = 1) \geq 0.6$.